SAT Math Mastery · Session 9

Right Triangles, Trigonometry and Circles

Time	3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.
Topics	right-triangle trigonometry (sin, cos, tan and cofunctions) · special right triangles and the Pythagorean theorem · altitude to the hypotenuse · Laws of Sines and Cosines, area with sin · distance and applied right triangles · circle equations (completing the square) · arcs, chords, tangents and inscribed angles · radians and degrees
Name / Date	_____ _____

PART A · IN CLASS — 36 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). In the Answers PDF, grey boxes show exactly what to type in Desmos; tap a grey dot to read an intersection, intercept or point. Type deg after an angle for degrees (for example sin(35 deg)); a letter such as k or b becomes a slider when Desmos offers one.

1. MEDIUM · DESMOS

If $\sin(2x)^\circ = \cos(x+15)^\circ$ and $0 < x < 45$, what is the value of x?

1. Type `y=sin(2x deg)` and `y=cos((x+15) deg)` (set the window to $0 < x < 45$).

2. Tap the intersection: (25, 0.766). The x-value is the answer.

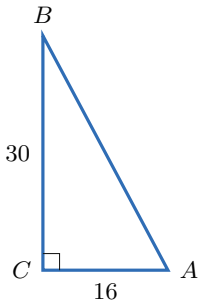
3. Type `sin(2*25 deg)` : it returns 0.766 and `cos((25+15) deg)` : it returns 0.766, so the two sides agree at $x = 25$.

4. The window excludes the other crossings such as $x = 105$, so $x = 25$.

Answer: 25

2. MEDIUM · NO CALCULATOR NEEDED

In right triangle ABC, the right angle is at C. If $AC = 16$ and $BC = 30$, what is $\tan A$?



(A) $\frac{16}{34}$

(B) $\frac{34}{16}$

(C) $\frac{8}{15}$

(D) $\frac{15}{8}$ ✓

1. $\tan A = \frac{\text{opposite}}{\text{adjacent}} = \frac{BC}{AC}$. Type `30/16` : it returns 1.875.

2. Type `15/8` : it returns 1.875, the same decimal, so option (D) matches.

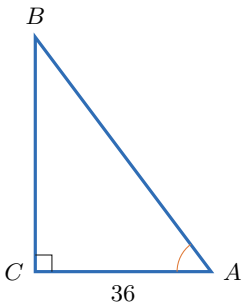
3. The other options: Type `8/15` : it returns 0.5333 (inverted) and `34/16` : it returns 2.125 (hypotenuse used).

4. So $\tan A = \frac{15}{8}$.

Answer: (D)

3. HARD · DESMOS

In right triangle ABC, the right angle is at C, and $\cos A = \frac{3}{5}$. If $AC = 36$, what is the length of AB?



1. $\cos A = \frac{AC}{AB}$, so $\frac{36}{AB} = \frac{3}{5} = 0.6$ with x standing for AB.

2. Type `y=36/x` and `y=0.6`.

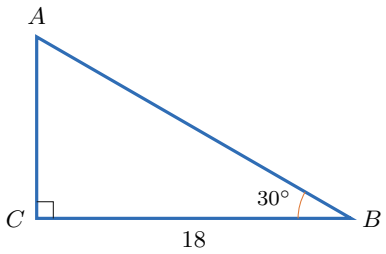
3. Tap the intersection: (60, 0.6), so $AB = 6$.

4. Check: Type `36/60`: it returns 0.6, which is $\frac{3}{5}$.

Answer: 60

4. MEDIUM · DESMOS

In triangle ABC, angle $C = 90°$, angle $B = 30°$, and $BC = 18$. What is the length of AC?



(A) 9

(B) $6\sqrt{3}$ ✓

(C) $9\sqrt{3}$

(D) $18\sqrt{3}$

1. AC is opposite the 30° angle, $BC = 18$ is adjacent, so $AC = 18\tan 30°$. Type `18tan(30 deg)`: it returns 10.3923.

2. Compare with the options: Type `6sqrt(3)`: it returns 10.3923 matches, so option (B).

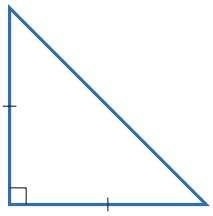
3. The others: Type `9sqrt(3)`: it returns 15.5885 and `18sqrt(3)`: it returns 31.1769 are too big, and 9 is just half of 18.

4. So $AC = 6\sqrt{3}$.

Answer: (B)

5. HARD · DESMOS

An isosceles right triangle has a perimeter of $40 + 20\sqrt{2}$. What is the length of the hypotenuse of the triangle?

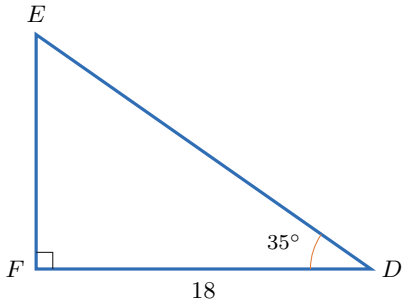


1. Let each leg be x. The hypotenuse is $x\sqrt{2}$, so the perimeter is $2x + x\sqrt{2}$.
2. Type `y=2x+x*sqrt(2)` and `y=40+20sqrt(2)`.
3. Tap the intersection: (20, 68.2843), so each leg is 2.
4. The hypotenuse is $x\sqrt{2}$: Type `20sqrt(2)`: it returns 28.2843, that is $20\sqrt{2}$.

Answer: $20\sqrt{2}$

6. HARD · DESMOS

In right triangle DEF, angle F is the right angle, angle D measures 35°, and side DF (adjacent to angle D) measures 18. Which of the following is closest to the length of side EF (opposite angle D)?



(A) 10.3

(B) 12.6 ✓

(C) 14.9

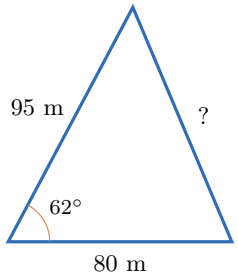
(D) 17.1

1. The legs are involved, so use tangent: $\tan 35° = \frac{EF}{18}$.
2. Type `18tan(35 deg)`: it returns 12.6037.
3. The closest option is 12.6, option (B).
4. Wrong-function answers differ: Type `18sin(35 deg)`: it returns 10.3244 (sine) and `18cos(35 deg)`: it returns 14.7447 (cosine).

Answer: (B)

7. HARD · DESMOS

A triangular plot of land has two sides measuring 80 meters and 95 meters, with an included angle of 62° between them. Which of the following is closest to the length, in meters, of the third side?



(A) 83.4

(B) 87.9

(C) 91.0 ✓

(D) 95.7

1. Two sides and the included angle: Law of Cosines, $c^2 = 80^2 + 95^2 - 2(80)(95)\cos 62°$.

2. Type `80^2+95^2-2*80*95cos(62 deg)` : it returns 8289.0322.

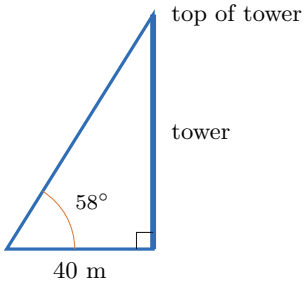
3. Type `sqrt(80^2+95^2-2*80*95cos(62 deg))` : it returns 91.0441.

4. The closest option is 91.0, option (C).

Answer: (C)

8. HARD · DESMOS

From a point on the ground 40 meters from the base of a communications tower, the angle of elevation to the top of the tower is 58°. Which of the following is closest to the height, in meters, of the tower?



(A) 33.9

(B) 47.1

(C) 64.0 ✓

(D) 75.5

1. The 40 m is adjacent to the 58° angle and the height is opposite: $h = 40\tan 58°$.

2. Type `40tan(58 deg)` : it returns 64.0134.

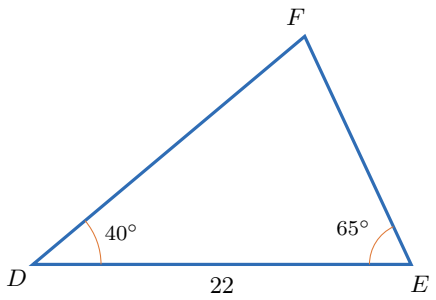
3. Closest option: 64.0, option (C).

4. A distractor comes from using sine instead of tangent: Type `40sin(58 deg)` : it returns 33.9219.

Answer: (C)

9. HARD · DESMOS

In triangle DEF, angle $D = 40°$, angle $E = 65°$, and side $DE = 22$. Which of the following is closest to the length of side EF (opposite angle D)?



(A) 12.6

(B) 14.6 ✓

(C) 16.9

(D) 19.4

1. Third angle: Type `180-40-65`: it returns 75 degrees.

2. Law of Sines: $EF = 22\dfrac{\sin 40°}{\sin 75°}$. Type `22sin(40 deg)/sin(75 deg)`: it returns 14.6402.

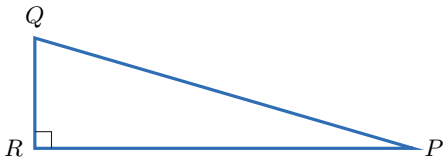
3. Closest option: 14.6, option (B).

4. Check using the other ratio: Type `22sin(65 deg)/sin(75 deg)`: it returns 20.6421 is side DF, not EF.

Answer: (B)

10. HARD · NO CALCULATOR NEEDED

In right triangle PQR, angle R is the right angle. If $\sin(P) = \dfrac{7}{25}$, what is the value of $\cos(Q)$?



(A) $\dfrac{7}{25}$ ✓

(B) $\dfrac{24}{25}$

(C) $\dfrac{25}{7}$

(D) $\dfrac{25}{24}$

1. The acute angles are complementary, so $\cos Q = \sin P = \dfrac{7}{25}$: Type `7/25`: it returns 0.28.

2. Check in Desmos: let $P = \arcsin\dfrac{7}{25}$ and type Type `cos(pi/2-arcsin(7/25))`: it returns 0.28.

3. Option (A) is $\dfrac{7}{25}$. The option $\dfrac{24}{25}$ is `24/25`: it returns 0.96, the cosine of P instead.

Answer: (A)

11. HARD · NO CALCULATOR NEEDED

Maya walked from Noah's house 4 miles due north, then 3 miles due east, while Noah walked 2 miles due south and 5 miles due west. If Noah were to walk directly to Maya's location, how many hours would it take him, if Noah walks at a rate of 5 miles per hour?

1. Origin at Noah's house: Maya ends at $(3, 4)$ and Noah at $(-5, -2)$.

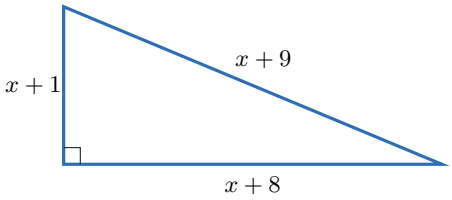
2. Distance: Type `sqrt((3-(-5))^2+(4-(-2))^2)` : it returns 10 miles.

3. Time: Type `10/5` : it returns 2 hours.

Answer: 2

12. HARD · DESMOS

The legs of a right triangle have lengths $x + 1$ and $x + 8$, and the hypotenuse has length $x + 9$. What is the area of the triangle?



1. Pythagorean theorem: $(x + 1)^2 + (x + 8)^2 = (x + 9)^2$. Type `y=(x+1)^2+(x+8)^2` and `y=(x+9)^2` .

2. Tap the intersection with a positive x: $(4, 169)$, so $x = 4$.

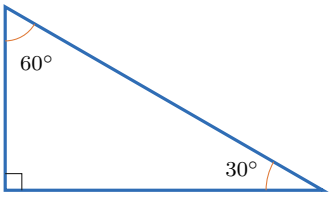
3. The legs are `4+1` : it returns 5 and `4+8` : it returns 12 (hypotenuse 13).

4. Area: Type `1/2*(4+1)(4+8)` : it returns 30.

Answer: 30

13. HARD · DESMOS

A 30°-60°-90° triangle has a perimeter of $18 + 6\sqrt{3}$. What is the length of its hypotenuse?



(A) 6

(B) $6\sqrt{3}$

(C) 12 ✓

(D) $12\sqrt{3}$

1. Sides are s, $s\sqrt{3}$, $2s$, so the perimeter is $3s + s\sqrt{3}$. Type `y=3x+x*sqrt(3)` and `y=18+6sqrt(3)`.

2. Tap the intersection: (6, 28.3923), so $s = 6$.

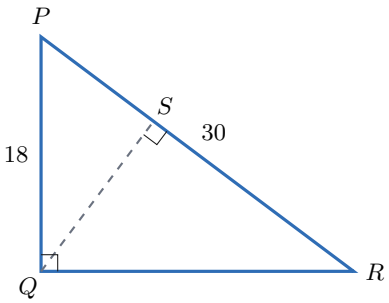
3. Hypotenuse $2s$: Type `2*6`: it returns 12.

4. Option (C).

Answer: (C)

14. HARD · DESMOS

In right triangle PQR, the right angle is at Q, $PQ = 18$, and $PR = 30$. Point S lies on $\overline{PR}$ such that $\overline{QS} \perp \overline{PR}$. What is the value of $\dfrac{QR}{QS}$?



1. $QR =$ `sqrt(30^2-18^2)`: it returns 24.

2. Area two ways: $QS = \dfrac{18 \cdot 24}{30}$. Type `18*24/30`: it returns 14.4.

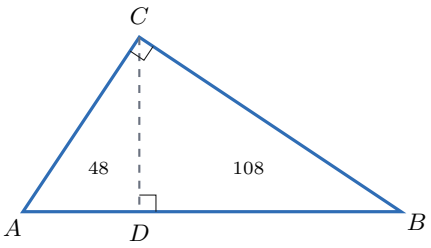
3. Ratio: Type `24/14.4`: it returns 1.6667.

4. That equals Type `5/3`: it returns 1.6667, so the answer is $\dfrac{5}{3}$.

Answer: $\dfrac{5}{3}$ **(or** 1.667)

15. HARD · DESMOS

In triangle ABC, the angle at C is a right angle. Point D lies on $\overline{AB}$ so that $\overline{CD} \perp \overline{AB}$. The area of triangle ACD is 48 and the area of triangle CDB is 108. What is the length of $\overline{AB}$?



1. Areas $48 : 108$ give $AD : DB = 4 : 9$: Type `48/108`: it returns 0.4444 and `4/9`: it returns 0.4444.

2. With $AD = 4k$, $DB = 9k$, $CD = 6k$, area of ACD is $12k^2$. Type `y=12x^2` and `y=48` (x stands for k).

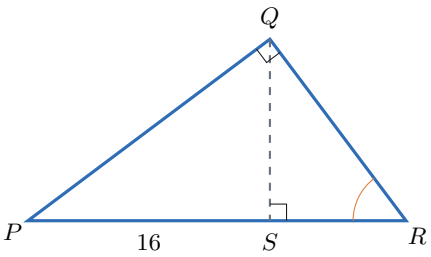
3. Tap the intersection: (2, 48), so $k = 2$.

4. $AB = 13k$: Type `13*2`: it returns 26.

Answer: 26

16. HARD · DESMOS

In triangle PQR, the angle at Q is a right angle. Point S lies on side $\overline{PR}$ so that $\overline{QS} \perp \overline{PR}$. If $\tan\angle QRP = \dfrac{4}{3}$ and $PS = 16$, what is the length of $\overline{SR}$?



(A) 12

(B) 15

(C) 25

(D) 9 ✓

1. Legs $PQ = 4k$, $QR = 3k$, $PR = 5k$, and $PQ^2 = PS \cdot PR$ gives $16k^2 = 16 \cdot 5k = 80k$.

2. Type `y=16x^2` and `y=80x` (x stands for k).

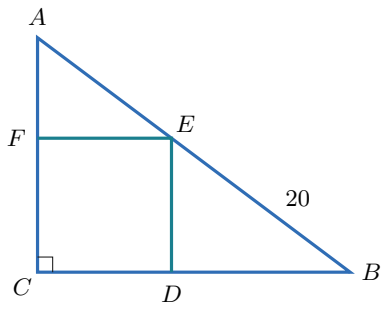
3. Tap the intersection with $x > 0$: (5, 400), so $k = 5$.

4. $PR =$ `5*5`: it returns 25 and $SR =$ `25-16`: it returns 9, option (D).

Answer: (D)

17. HARD · DESMOS

In right triangle ABC, $\angle C = 90°$, $AB = 20$, and the area of the triangle is 96. Square $CDEF$ is inscribed in the triangle, with D on $\overline{BC}$, F on $\overline{AC}$, and E on $\overline{AB}$. What is the side length of the square?



(A) $\frac{48}{7}$ ✓

(B) $\frac{24}{7}$

(C) $\frac{48}{5}$

(D) $\frac{96}{7}$

1. Let the legs be a, b: $ab = 2 \cdot 96$ and $a^2 + b^2 = 400$. Type `2*96`: it returns 192.

2. $a + b = \sqrt{400 + 2ab}$: Type `sqrt(400+2*192)`: it returns 28.

3. Square side $s = \frac{ab}{a+b}$: Type `192/28`: it returns 6.8571.

4. Type `48/7`: it returns 6.8571, the same, so option (A).

Answer: (A)

18. HARD · DESMOS

In right triangle PQR, $\angle R = 90°$ and $\sin P + \sin Q = \frac{\sqrt{6}}{2}$. What is the value of $\tan P + \tan Q$?

1. $\sin Q = \cos P$, so the condition is $\sin P + \cos P = \frac{\sqrt{6}}{2}$ with x standing for P in degrees. Type `y=sin(x deg)+cos(x deg)` and `y=sqrt(6)/2`.

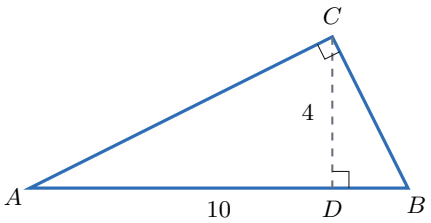
2. Tap the first intersection: (15, 1.2247), so $P = 15°$ and $Q = 75°$ (the other crossing swaps them).

3. Type `tan(15 deg)+tan(75 deg)`: it returns 4.

Answer: 4

19. HARD · DESMOS

In right triangle ABC, the right angle is at C and $AB = 10$. Point D lies on $\overline{AB}$ with $\overline{CD} \perp \overline{AB}$ and $CD = 4$. If $AD > DB$, what is the value of $\cos A$?



(A) $\dfrac{\sqrt{5}}{5}$

(B) $\dfrac{4}{5}$

(C) $\dfrac{1}{2}$

(D) $\dfrac{2\sqrt{5}}{5}$ ✓

1. With $AD = x$, $DB = 10 - x$ and $CD^2 = AD \cdot DB = 16$. Type `y=x(10-x)` and `y=16`.

2. Tap the intersection with the larger x (because $AD > DB$): (8, 16), so $AD = 8$.

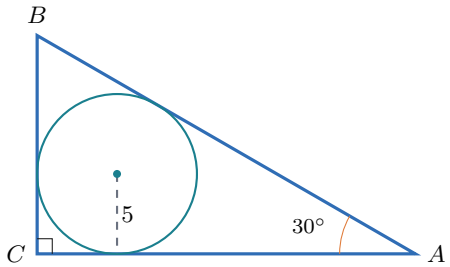
3. $AC =$ `sqrt(8^2+4^2)`: it returns 8.9443 and $\cos A = \dfrac{AD}{AC}$: Type `8/sqrt(80)`: it returns 0.8944.

4. Type `2sqrt(5)/5`: it returns 0.8944, the same, so option (D).

Answer: (D)

20. HARD · DESMOS

In triangle ABC, angle C is a right angle and angle A measures 30°. The circle inscribed in the triangle has radius 5. What is the length of $\overline{AC}$?



(A) $15 + 5\sqrt{3}$ ✓

(B) $10 + 10\sqrt{3}$

(C) $10\sqrt{3} - 10$

(D) $5 + 5\sqrt{3}$

1. With $BC = x$: $AC = x\sqrt{3}$, $AB = 2x$, and $r = \dfrac{x + x\sqrt{3} - 2x}{2} = \dfrac{x(\sqrt{3} - 1)}{2}$. Type `y=x(sqrt(3)-1)/2` and `y=5`.

2. Tap the intersection: (13.6603, 5), so $BC \approx 13.6603$.

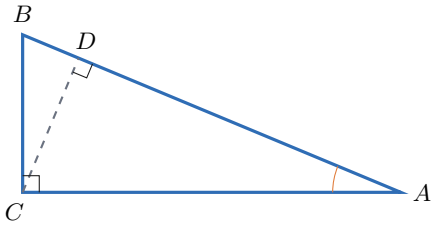
3. $AC = x\sqrt{3}$: Type `10/(sqrt(3)-1)*sqrt(3)`: it returns 23.6603.

4. Type `15+5sqrt(3)`: it returns 23.6603, the same, so option (A).

Answer: (A)

21. HARD · DESMOS

In right triangle ABC, the right angle is at C and $\tan A = \dfrac{5}{12}$. The area of the triangle is 120. Point D lies on $\overline{AB}$ such that $\overline{CD} \perp \overline{AB}$. What is the length of $\overline{AD}$?



(A) $\dfrac{120}{13}$

(B) 24

(C) $\dfrac{50}{13}$

(D) $\dfrac{288}{13}$ ✓

1. Sides $5k$, $12k$, $13k$, so the area is $30k^2$. Type `y=30x^2` and `y=120` (x stands for k).

2. Tap the intersection: (2, 120), so $k = 2$, $AC = 24$ and $AB = 26$.

3. Leg rule $AD = \dfrac{AC^2}{AB}$: Type `24^2/26`: it returns 22.1538.

4. Type `288/13`: it returns 22.1538, the same, so option (D).

Answer: (D)

22. HARD · DESMOS

For $0 < x < 45$, the equation

$$\sin\left((5x - 20)^\circ\right) = \cos\left((x + 14)^\circ\right)$$

has more than one solution. What is the sum of all solutions x in this interval?

1. Type `y=sin((5x-20) deg)` and `y=cos((x+14) deg)`, with the window $0 < x < 45$.

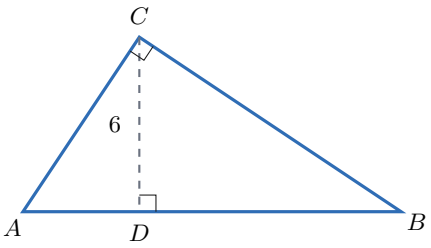
2. Tap both intersections: (16, 0.866) and (31, 0.7071).

3. The solutions are $x = 16$ and $x = 31$. Sum: Type `16+31`: it returns 47.

Answer: 47

23. HARD · DESMOS

In right triangle ABC, the angle at C is a right angle. Point D lies on $\overline{AB}$ so that $\overline{CD} \perp \overline{AB}$. The length of $\overline{CD}$ is 6, and DB is 5 units longer than AD. What is the value of $\sin B$?



(A) $\dfrac{2\sqrt{13}}{13}$ ✓

(B) $\dfrac{3\sqrt{13}}{13}$

(C) $\dfrac{4}{13}$

(D) $\dfrac{6}{13}$

1. With $AD = x$ and $DB = x + 5$: $CD^2 = AD \cdot DB = 36$. Type `y=x(x+5)` and `y=36`.

2. Tap the intersection with $x > 0$: (4, 36), so $AD = 4$, $DB = 9$, $AB = 13$.

3. $AC = \sqrt{4^2 + 6^2}$ and $\sin B = \dfrac{AC}{AB}$: Type `sqrt(4^2+6^2)/13`: it returns 0.5547.

4. Type `2sqrt(13)/13`: it returns 0.5547, the same, so option (A).

Answer: (A)

24. HARD · DESMOS

In triangle PQR, the measure of angle P is 30° and $PR = 14$. The length $QR = a$ is a positive integer with $a \leq 20$. For how many values of a does the given information determine exactly one triangle, up to congruence?

1. Height from R to line PQ: Type `14sin(30 deg)`: it returns 7. So $a < 7$ gives no triangle, $a = 7$ one, $7 < a < 14$ two, and $a \geq 14$ one.

2. Type the values that give exactly one triangle as a list: `L=[7,14...20]`.

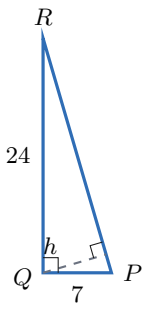
3. Type `L.length`: it returns 8.

4. So the count is 8.

Answer: 8

25. HARD · DESMOS

In right triangle PQR, the right angle is at Q, $PQ = 7$ and $QR = 24$. The altitude from Q to hypotenuse $\overline{PR}$ has length h. What is the value of h? (Enter as a decimal.)



1. Hypotenuse: Type `sqrt(7^2+24^2)` : it returns 25.

2. Area two ways: $h = \dfrac{7 \cdot 24}{25}$. Type `7*24/25` : it returns 6.72.

3. Check: Type `1/2*25*6.72` : it returns 84 equals `1/2*7*24` : it returns 84, the area of the triangle.

Answer: 6.72

26. HARD · DESMOS

In a right triangle, angle θ is acute and $\sin\theta = 0.6$. What is the value of $\cos(90° - \theta) + \tan\theta$?

(A) 1.35 ✓ (B) 1.6

(C) 0.75 (D) 1.2

1. $\cos(90° - \theta) = \sin\theta = 0.6$ and $\tan\theta$: Type `tan(arcsin(0.6))` : it returns 0.75.

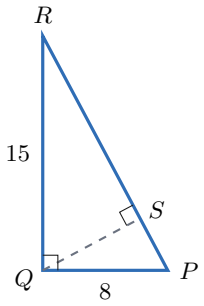
2. Sum: Type `0.6+tan(arcsin(0.6))` : it returns 1.35.

3. Check directly: Type `cos(pi/2-arcsin(0.6))+tan(arcsin(0.6))` : it returns 1.35, option (A).

Answer: (A)

27. HARD · DESMOS

In right triangle PQR, the right angle is at Q, $PQ = 8$ and $QR = 15$. Point S is on $\overline{PR}$ such that $\overline{QS}$ is perpendicular to $\overline{PR}$. What is the length of $\overline{QS}$? (Enter as a fraction.)



1. Hypotenuse: Type `sqrt(8^2+15^2)` : it returns 17.

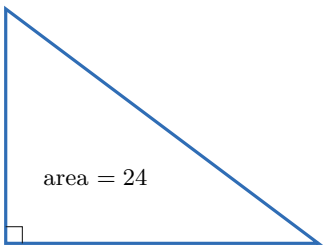
2. Area two ways: $QS = \frac{8 \cdot 15}{17}$: Type `8*15/17` : it returns 7.0588.

3. Type `120/17` : it returns 7.0588, the same fraction, $\frac{120}{17}$.

Answer: $\frac{120}{17}$

28. HARD · DESMOS

In a right triangle, the sine of one acute angle is $\frac{3}{5}$ and the area of the triangle is 24. What is the length of the hypotenuse?



1. Sides $3k, 4k, 5k$ give area $6k^2$. Type `y=6x^2` and `y=24` (x stands for k).

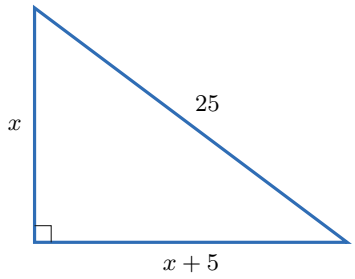
2. Tap the intersection: (2, 24), so $k = 2$.

3. Hypotenuse $5k$: Type `5*2` : it returns 10.

Answer: 10

29. HARD · DESMOS

The legs of a right triangle have lengths x and $x + 5$, and the hypotenuse is 25. What is the area of the triangle?



1. Pythagorean theorem: $x^2 + (x + 5)^2 = 25^2$. Type `y=x^2+(x+5)^2` and `y=625`.
2. Tap the intersection with a positive x: (15, 625), so $x = 15$ (the other intersection has $x = -20$).
3. Legs 15 and 20: area Type `1/2*15*20`: it returns 150.

Answer: 150

30. HARD · DESMOS

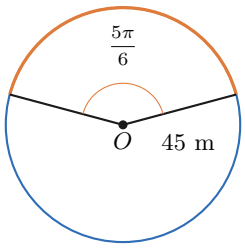
In the xy-plane, the graph of $x^2 + y^2 + 4x - 10y + k = 0$ is a circle with radius 3. What is the value of k?

1. Completing the square gives $(x + 2)^2 + (y - 5)^2 = 29 - k$. Set the radius squared to 9: type `y=29-x` and `y=9` (x stands for k).
2. Tap the intersection: (20, 9), so $k = 20$.
3. Check: type `x^2+y^2+4x-10y+20=0`. Its top and bottom points $(-2, 8)$ and $(-2, 2)$ are 6 apart, so the radius is `(8-2)/2`: it returns 3.

Answer: 20

31. HARD · DESMOS

A circular plaza has a radius of 45 meters. A decorative walking path forms an arc along the plaza's boundary that subtends a central angle of $\frac{5\pi}{6}$ radians. What is the length of the arc, in meters?



(A) $\frac{25\pi}{2}$

(B) $\frac{75\pi}{2}$ ✓

(C) 45π

(D) 150π

1. Arc length $s = r\theta$ (radians): Type `45*5pi/6` : it returns 117.8097.

2. Type `75pi/2` : it returns 117.8097, the same, so option (B).

3. The option 45π is `45pi` : it returns 141.3717, which uses $\theta = \pi$.

Answer: (B)

32. HARD · DESMOS

A circle in the xy-plane has the equation $x^2 + y^2 - 8x + 2y - 19 = 0$. What is the diameter of the circle?

1. Type the circle: `x^2+y^2-8x+2y-19=0` . Tap its rightmost and leftmost points $(10, -1)$ and $(-2, -1)$.

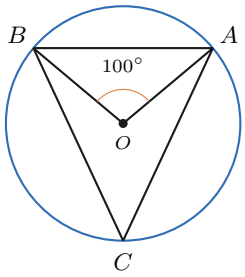
2. Diameter is the width: Type `10-(-2)` : it returns 12.

3. Check with the completed square $(x-4)^2 + (y+1)^2 = 36$: Type `2sqrt(36)` : it returns 12.

Answer: 12

33. HARD · DESMOS

In the circle above, chord AB subtends a central angle of $100°$ at center O. Point C lies on the major arc. What is the measure of angle ACB, in degrees?



1. The inscribed angle is half the central angle on the same arc: Type `100/2` : it returns 50.

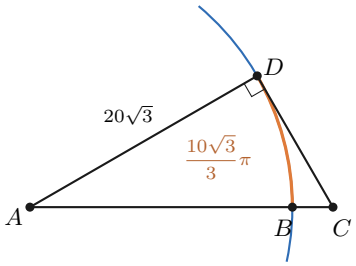
2. Check the other direction: Type `2*50` : it returns 100, the central angle.

3. So $\angle ACB = 50°$.

Answer: 50

34. HARD · DESMOS

A circle has center A and radius $20\sqrt{3}$. Segment CD is tangent to the circle at point D, so triangle ACD has a right angle at D. Point B lies on the circle along ray AC, and minor arc BD has length $\dfrac{10\sqrt{3}}{3}\pi$. What is the length of CD?



1. Fraction of the circle: Type `(10sqrt(3)/3*pi)/(2pi*20sqrt(3))` : it returns 0.0833.

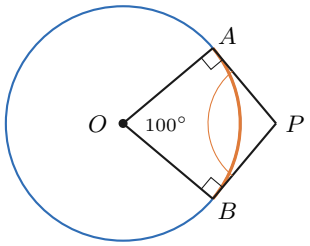
2. Central angle: Type `360*(10sqrt(3)/3*pi)/(2pi*20sqrt(3))` : it returns 30 degrees.

3. $CD = AD\tan 30°$: Type `20sqrt(3)*tan(30 deg)` : it returns 20.

Answer: 20

35. HARD · DESMOS

In a figure, PA and PB are tangent to a circle centered at O, at points A and B respectively, so $OA \perp PA$ and $OB \perp PB$. If $\angle APB = 100°$ and the circle has a circumference of 36π, what is the length of minor arc AB?



(A) 6π

(B) 8π ✓

(C) 9π

(D) 10π

1. Central angle: Type `360-90-90-100`: it returns 80 degrees.

2. Radius is 18, arc length: Type `80/360*36pi`: it returns 25.1327.

3. Type `8pi`: it returns 25.1327, the same, so option (B).

Answer: (B)

36. HARD · DESMOS

A circle has equation $(x+4)^2 + (y-b)^2 = 100$, where b is a constant. The points $(4, 13)$ and $(-10, -1)$ both lie on the circle. What is the value of b?

1. Type the circle with a slider: `(x+4)^2+(y-b)^2=100` (accept the slider for b), and plot `(4,13)` and `(-10,-1)`.

2. Slide b until both points sit on the circle: $b = 7$ (the equation with $b = 7$ is `(x+4)^2+(y-7)^2=100`).

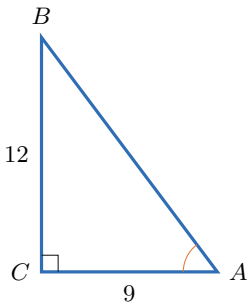
3. Check: Type `(4+4)^2+(13-7)^2`: it returns 100 and `(-10+4)^2+(-1-7)^2`: it returns 100.

Answer: 7

PART B · SELF-ASSESSMENT — 21 PROBLEMS

37. MEDIUM · NO CALCULATOR NEEDED

In right triangle ABC, angle C is the right angle, $AC = 9$, and $BC = 12$. What is $\sin(A)$?



(A) $\frac{4}{5}$ ✓

(B) $\frac{3}{4}$

(C) $\frac{4}{3}$

(D) $\frac{3}{5}$

1. Hypotenuse: Type `sqrt(9^2+12^2)` : it returns 15.

2. $\sin A = \frac{BC}{AB}$: Type `12/15` : it returns 0.8.

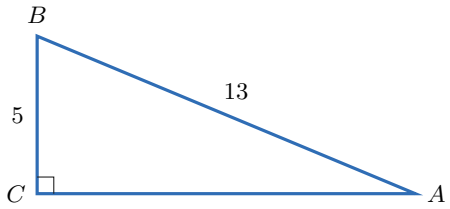
3. Type `4/5` : it returns 0.8, the same, so option (A).

4. The option $\frac{3}{5}$ is `9/15` : it returns 0.6, which is $\cos A$.

Answer: (A)

38. MEDIUM · NO CALCULATOR NEEDED

In right triangle ABC, angle C is the right angle, $AB = 13$ (the hypotenuse), and $BC = 5$. What is $\cos(A)$?



(A) $\frac{5}{12}$

(B) $\frac{12}{5}$

(C) $\frac{5}{13}$

(D) $\frac{12}{13}$ ✓

1. Missing leg: Type `sqrt(13^2-5^2)` : it returns 12.

2. $\cos A = \frac{AC}{AB}$: Type `12/13` : it returns 0.9231.

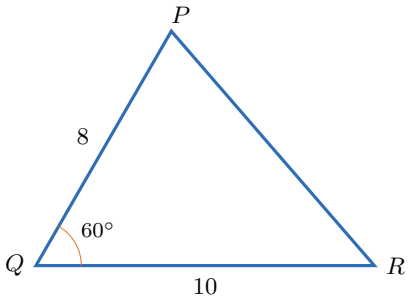
3. Compare: Type `5/13` : it returns 0.3846 is $\sin A$, not the answer; option (D) has $\frac{12}{13}$.

4. So $\cos A = \frac{12}{13}$.

Answer: (D)

39. Medium · Desmos

In $\triangle PQR$, $PQ = 8$, $QR = 10$, and $m\angle Q = 60°$. What is the area of $\triangle PQR$?



(A) 20 (B) $20\sqrt{2}$

(C) $20\sqrt{3}$ ✓ (D) $40\sqrt{3}$

1. Area from two sides and the angle between them: $\frac{1}{2}(8)(10)\sin 60°$.

2. Type `1/2*8*10*sin(60 deg)` : it returns 34.641.

3. Type `20sqrt(3)` : it returns 34.641, the same, so option (C).

4. The others: Type `20sqrt(2)` : it returns 28.2843 and `40sqrt(3)` : it returns 69.282 do not match.

Answer: (C)

40. Medium · No calculator needed

What is the distance between the points $(-3, -9)$ and $(9, -4)$?

1. Differences: $9 - (-3) = 12$ and $-4 - (-9) = 5$.

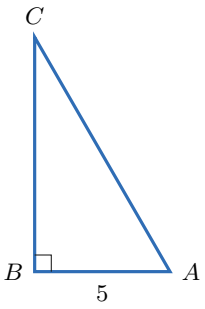
2. Type `sqrt((9-(-3))^2+(-4-(-9))^2)` : it returns 13.

3. So the distance is 13.

Answer: 13

41. Medium · Desmos

In triangle ABC, angle B is a right angle and the measure of angle A is twice the measure of angle C. If $AB = 5$, what is the length of BC?



(A) 5

(B) $5\sqrt{2}$

(C) $5\sqrt{3}$ ✓

(D) 10

1. With $x = BC$: angle $A = \arctan\frac{x}{5}$ and angle $C = \arctan\frac{5}{x}$. The condition is $A = 2C$. Type `y=arctan(x/5)` and `y=2arctan(5/x)`.

2. Tap the intersection: (8.6603, 1.0472), so $BC \approx 8.6603$.

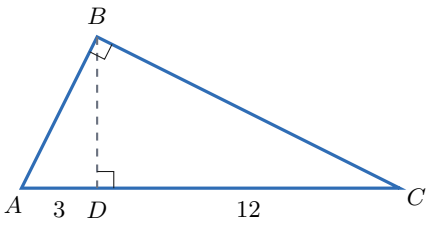
3. Compare with the options: Type `5sqrt(3)`: it returns 8.6603, the same, so option (C).

4. Check: Type `arctan(5sqrt(3)/5)*180/pi`: it returns 60 degrees for A and `arctan(5/(5sqrt(3)))*180/pi`: it returns 30 degrees for C.

Answer: (C)

42. Medium · Desmos

In right triangle ABC, the right angle is at B, and BD is the altitude drawn to hypotenuse AC (with D on AC). If $AD = 3$ and $DC = 12$, what is the length of BD?



(A) 4

(B) 6 ✓

(C) 7.5

(D) 9

1. Altitude rule: $BD^2 = AD \cdot DC$. Type `3*12`: it returns 36.

2. Type `sqrt(3*12)`: it returns 6, so $BD = 6$, option (B).

3. Check: Type `(3^2+6^2)+(12^2+6^2)`: it returns 225 equals `15^2`: it returns 225, so the triangle is right-angled at B.

Answer: (B)

43. Medium · Desmos

$$\sin(x^\circ) = \cos\left((2x+9)^\circ\right)$$

In the equation above, x is between 0 and 30. What is the value of x?

1. Type `y=sin(x deg)` and `y=cos((2x+9) deg)` (window $0 < x < 30$).

2. Tap the intersection: (27, 0.454).

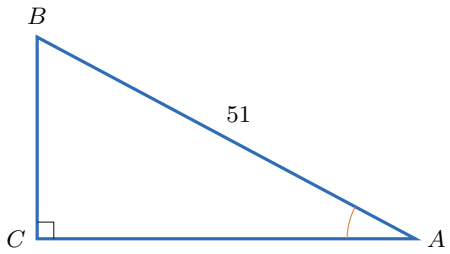
3. Check the angles are complementary: Type `27+(2*27+9)` : it returns 90.

4. So $x = 27$.

Answer: 27

44. Medium · Desmos

In right triangle ABC, the right angle is at C, $\tan A = \dfrac{8}{15}$, and $AB = 51$. What is the perimeter of triangle ABC?



1. Sides $8k$, $15k$ and hypotenuse $\sqrt{(8k)^2 + (15k)^2}$ (x stands for k). Type `y=sqrt((8x)^2+(15x)^2)` and `y=51` .

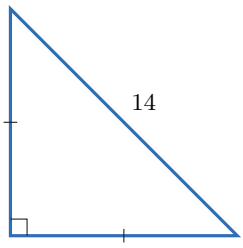
2. Tap the intersection: (3, 51), so $k = 3$.

3. Perimeter: Type `8*3+15*3+51` : it returns 120.

Answer: 120

45. Medium · Desmos

An isosceles right triangle has a hypotenuse of length 14. What is the perimeter of the triangle?



(A) $14 + 7\sqrt{3}$

(B) 28

(C) $14 + 7\sqrt{2}$

(D) $14 + 14\sqrt{2}$ ✓

1. Each leg is $\dfrac{14}{\sqrt{2}}$: Type `14/sqrt(2)` : it returns 9.8995.

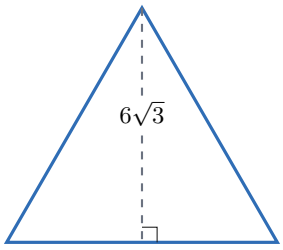
2. Perimeter: Type `14+2*14/sqrt(2)` : it returns 33.799.

3. Type `14+14sqrt(2)` : it returns 33.799, the same, so option (D).

Answer: (D)

46. Medium · Desmos

The height of an equilateral triangle is $6\sqrt{3}$. What is the perimeter of the triangle?



1. Height of an equilateral triangle with side x is $\frac{x\sqrt{3}}{2}$. Type `y=x*sqrt(3)/2` and `y=6sqrt(3)`.

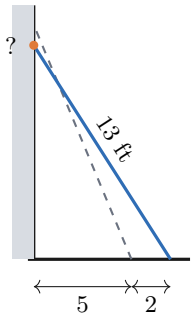
2. Tap the intersection: (12, 10.3923), so $s = 12$.

3. Perimeter: Type `3*12`: it returns 36.

Answer: 36

47. Medium · Desmos

A 13-foot ladder leans against a vertical wall. Its base is 5 feet from the wall. The base is then pulled 2 feet farther from the wall, so the ladder is 7 feet from the wall. To the nearest tenth of a foot, how high up the wall does the ladder reach now?



1. Height $= \sqrt{13^2 - 7^2}$: Type `sqrt(13^2-7^2)`: it returns 10.9545.

2. To the nearest tenth: Type `round(sqrt(13^2-7^2),1)`: it returns 11 (that is 11.0).

3. Check how far the top slid down: Type `12-sqrt(13^2-7^2)`: it returns 1.0455 feet (the ladder started 12 feet up).

Answer: 11.0

48. MEDIUM · NO CALCULATOR NEEDED

What is the value of $\cos\dfrac{5\pi}{3}$?

(A) $-\dfrac{\sqrt{3}}{2}$ (B) $-\dfrac{1}{2}$

(C) $\dfrac{\sqrt{3}}{2}$ **(D) $\dfrac{1}{2}$ ✓**

1. Type `cos(5pi/3)` : it returns 0.5 (Desmos is in radian mode, which is what we want here).

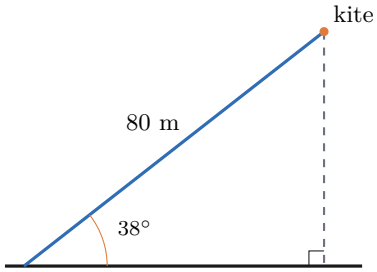
2. Type `1/2` : it returns 0.5, the same, so option (D).

3. The options with $\dfrac{\sqrt{3}}{2}$ are `sqrt(3)/2` : it returns 0.866, which is $\cos\dfrac{\pi}{6}$, so they are wrong.

Answer: (D)

49. MEDIUM · DESMOS

A kite is flying at the end of a straight string that is 80 meters long. The string makes an angle of 38° with level ground. To the nearest meter, how high above the ground is the kite?



1. Height is opposite the 38° angle and the string is the hypotenuse: $h = 80\sin 38°$.

2. Type `80sin(38 deg)` : it returns 49.2529.

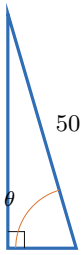
3. To the nearest meter: 49.

4. Check the tangent mistake: Type `80tan(38 deg)` : it returns 62.5029 is too large.

Answer: 49

50. Medium · Desmos

In a right triangle, the hypotenuse is 50 and the cosine of an acute angle θ is $\frac{7}{25}$. What is the area of the triangle?



1. Adjacent leg: Type `50*7/25` : it returns 14.

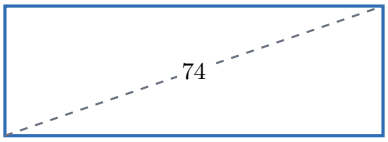
2. Other leg: Type `sqrt(50^2-14^2)` : it returns 48.

3. Area: Type `1/2*14*48` : it returns 336.

Answer: 336

51. Medium · Desmos

The diagonal of a rectangle is 74 and the ratio of its two side lengths is $12 : 35$. What is the area of the rectangle?



1. Sides $12k$ and $35k$ (x stands for k): the diagonal is $\sqrt{(12k)^2 + (35k)^2}$. Type `y=sqrt((12x)^2+(35x)^2)` and `y=74` .

2. Tap the intersection: (2, 74), so $k = 2$.

3. Area: Type `(12*2)(35*2)` : it returns 1680.

Answer: 1680

52. Medium · Desmos

A circle in the xy-plane is defined by the equation $x^2 + y^2 - 6x + 8y = 0$. What is the radius of the circle?

(A) 3 (B) 4

(C) 5 ✓ (D) 25

1. Type the circle: `x^2+y^2-6x+8y=0` . Tap its intercepts (6, 0), (0, −8) and (0, 0).

2. Type the completed-square form `(x-3)^2+(y+4)^2=25` : the same circle (it passes through the same three points).

3. Radius: Type `sqrt(25)` : it returns 5. Option (C).

Answer: (C)

53. MEDIUM · DESMOS

A circle in the xy-plane is defined by the equation $x^2 + y^2 + 10x - 4y + 13 = 0$. What is the radius of the circle?

(A) 4 ✓ (B) 8

(C) 13 (D) 16

1. Type the circle: `x^2+y^2+10x-4y+13=0`. Tap its rightmost and leftmost points $(-1, 2)$ and $(-9, 2)$.

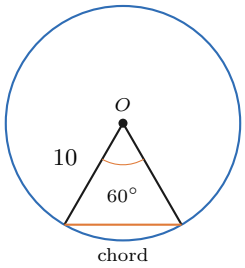
2. Radius is half the width: Type `(-1-(-9))/2`: it returns 4.

3. Option (A).

Answer: (A)

54. MEDIUM · DESMOS

In a circle with radius 10, a chord subtends a central angle of 60°. What is the length of the chord?



(A) 5 (B) $5\sqrt{3}$

(C) 10 ✓ (D) $10\sqrt{3}$

1. Chord length $= 2r \sin\frac{\theta}{2}$: Type `2*10*sin(30 deg)`: it returns 10.

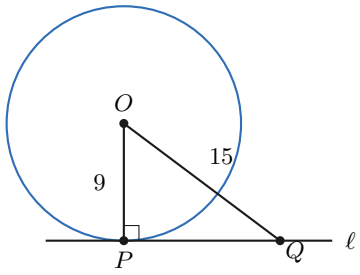
2. Option (C) is 10.

3. Check an option: Type `10sqrt(3)`: it returns 17.3205 would need a 120° angle, since `2*10*sin(60 deg)`: it returns 17.3205.

Answer: (C)

55. Medium · Desmos

In the figure, line ℓ is tangent to the circle with center O at point P. If $OP = 9$ and $OQ = 15$ for a point Q on line ℓ, what is the length of PQ?



(A) 6

(B) 12 ✓

(C) $\sqrt{306}$

(D) 18

1. The tangent is perpendicular to the radius, so $PQ = \sqrt{OQ^2 - OP^2}$: Type `sqrt(15^2-9^2)` : it returns 12.

2. Option (B) is 12.

3. The option $\sqrt{306}$ is `sqrt(15^2+9^2)` : it returns 17.4929, which adds instead of subtracts.

Answer: (B)

56. Medium · No calculator needed

An angle has a measure of $\dfrac{7\pi}{12}$ radians. What is the measure of the angle, in degrees?

1. Multiply by $\dfrac{180}{\pi}$: Type `7pi/12*180/pi` : it returns 105.

2. So the angle is 105°.

3. Check the reverse: Type `105*pi/180/pi*12` : it returns 7 (that is 7 twelfths of π).

Answer: 105

57. Medium · No calculator needed

The number of radians in a 540-degree angle can be written as $a\pi$, where a is a constant. What is the value of a?

1. Multiply by $\dfrac{\pi}{180}$ and divide by π: Type `540*pi/180/pi` : it returns 3.

2. So $540° = 3\pi$ and $a = 3$.

3. Check: Type `3pi*180/pi` : it returns 540 degrees.

Answer: 3